NAME: _____ Geometry (FL B.E.S.T.)

10.4 Cross-Sections

A **cross-section** is the 2D shape formed when a plane slices through a 3D solid. The shape you get depends entirely on the *orientation* of the cutting plane: parallel to a base, through an axis, through an apex, tilted, or through several faces at once. Beyond identifying the shape, this worksheet computes actual cross-section *dimensions and areas* — for a cone or pyramid cut parallel to the base, the cross-section is **similar** to the base, scaled by the ratio of distances from the apex (a direct application of similar triangles); for a sphere cut off-center, the Pythagorean Theorem relates the sphere's radius, the distance to the center, and the smaller circle's radius.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Cylinder cut parallel to its base. Name the cross-section.

2. Cylinder radius 4, height 9, cut through its axis. Find the cross-section's dimensions and area.

3. Cone radius 9, height 12, cut parallel to the base at 8 units from the apex. Find the cross-section's radius and area.

4. The same cone (radius 9, height 12) cut through the apex and axis. Find the cross-section's shape and area.

5. Sphere radius 10, cut through the center. Find the cross-section's area.

6. Sphere radius 17, cut 8 units from the center. Find the cross-section's radius and area.

7. Square pyramid, base edge 10, height 15, cut parallel to the base at 9 units from the apex. Find the cross-section's edge and area.

8. Cube edge 8, corner sliced off through the three adjacent vertices. Find the cross-section's side length and area.

9. Cube edge 7, sliced through four vertices (not a corner). Find the cross-section's shape and area.

10. Cylinder radius 5, sliced by a plane tilted 45° from horizontal. Find the ellipse's semi-minor and semi-major axes.

11. Triangular prism cut parallel to its triangular base. Name the cross-section.

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12. Define "great circle."

13. Under what condition does a cone's cross-section become an ellipse instead of a circle or triangle?

14. A cylindrical log of radius 6 in is cut by a plane parallel to its ends. Name the cross-section and find its area.

15. A cone has base radius 10 and height 24. A plane parallel to the base cuts the cone so the circular cross-section has area 25π. How far from the apex is the cutting plane?

✓ Answer Key

1. Circle (congruent to the base)
2. Rectangle 8×9; area 72
3. $r = 6$; area 36π
4. Isosceles triangle, base 18, height 12; area 108
5. 100π
6. $r = 15$; area 225π
7. edge 6; area 36
8. side $8\sqrt{2}$; area $32\sqrt{3}$
9. Rectangle $7 \times 7\sqrt{2}$; area $49\sqrt{2}$
10. semi-minor 5; semi-major $5\sqrt{2}$
11. Triangle congruent to the base
12. The circular cross-section of a sphere through its center (largest possible)
13. When the cutting plane is tilted (not parallel to the base and not through the apex/axis)
14. Circle; area 36π in^2
15. 12 units from the apex